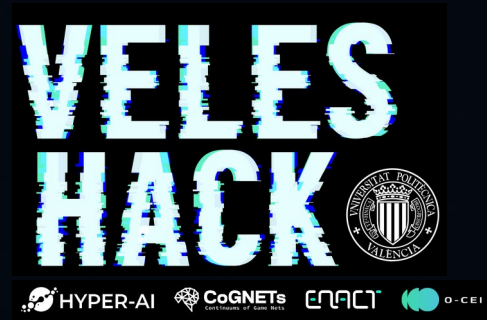


TangleLens

IOTA Advanced Explorer for Eclipse aeriOS · Challenge 2 (O-CEI)



Funded by
the European Union

GitHub repo

- **GitHub repo:**
<https://github.com/moe-a-m/tanglelens>
- **Licence:** Apache-2.0, including modified Eclipse aeriOS code (see NOTICE)
- **Run it:** start the aeriOS tangle (iota-tangle/), then make up · make demo · make test
- **Evidence:** raw Hornet responses, test summaries and clean-clone runs in reports/
- **Stack:** Python, FastAPI, Flask, PostgreSQL, Mosquitto MQTT, HORNET 2.0.2, Docker Compose

TangleLens IOTA Advanced Explorer for Eclipse aeriOS Hornet **HORNET** reachable, latest milestone **15012**

Search message text: e.g. ie-2, be_creative Block ID or prefix: 8x... Tag: trust.score or trust From (UTC): mm/dd/yyyy, --:--:-- To (UTC): mm/dd/yyyy, --:--:-- Source: aeriOS/trust-manager Trace: domain-1-ie-2

Search **Clear**

All 5 **Confirmed 4** **Content mismatch 1**

2 open alerts: critical events or integrity failures

Status	Tag and source	Submitted (UTC)	Block
Confirmed	self.requestration	2026-10-07 12:14:46	@e4419a8f... b2c6e2
Content mismatch	trust.score aeriOS/trust-manager - trace domain-1-ie-2-151445	2026-10-07 12:14:46	@x26b2ef33... 7d269d
Confirmed	trust.security.alert aeriOS/self-security - trace domain-1-ie-2-151445	2026-10-07 12:14:46	@x548c1228... b67b51
Confirmed	trust.score aeriOS/trust-manager - trace domain-1-ie-1-151445	2026-10-07 12:14:46	@xafa00f52... 7d32c3
Confirmed	trust.reliability aeriOS/self-awareness - trace domain-1-ie-1-151445	2026-10-07 12:14:46	@xcdd32b2... 8a2391

1-5 of 5 **Previous** **Next**

The Tangle holds exactly this message.
Referenced by milestone 15010 at 2026-10-07 12:14:46 UTC.

✓ Block exists on node ✓ Block is solid ✓ Referenced by milestone
✓ Tag matches ✓ Message content matches

Verify again **Copy block ID**

Message

```
{
  "domain_name": "domain-1",
  "ie": "ie-1",
  "cpucycles": 0,
  "currentcpusage": 37.5,
  "ramcapacity": 10384,
  "availableram": 9120,
  "currentramusage": 44.3
}
```

Details

Block ID: @xcdd32b28a5575d90e8f3747a8b3dfb4326368a7e1bd8db5fd86fb6d288a2391
Tag: trust.reliability
Type: -
Source: aeriOS/self-awareness

Summary

- **What:** a searchable, continuously verified copy of every message Eclipse aeriOS writes to its private IOTA Tangle. The Tangle stays the authority.
- **App 1, extended aeriOS Messages API:** same /upload contract and HTTP 200; after Hornet accepts a block it forwards the block id and the exact bytes sent, over HTTP and MQTT.
- **App 2, Advanced Explorer:** PostgreSQL copy enriched with readable metadata; REST API and web UI search by block id, date and tag, plus source, text, status and trace.
- **Validation:** every block is checked with Hornet's GET block metadata (solid, milestone-referenced, conflicting) and GET block (content compared as hex, decoded JSON and SHA-256).
- **Audit:** append-only history of every check; all messages re-audited every 5 minutes, so later tampering is caught.
- **Runs on the real aeriOS private tangle** (HORNET 2.0.2) with Docker Compose; tested from a fresh clone.

Highlights

- **Tampering is caught:** an IE's trust score inflated from 0.61 to 0.95 in the database is flagged, shown next to the Tangle copy and alerted.
- **Solid is not confirmed:** 8 explicit states; measured on the real node: solid within 20 ms, milestone median 3.3 s.
- **Beyond the brief (UPV ideas #1, #3, #4):** trace timelines, alerts (UI, webhook, MQTT), MQTT live feed.
- **Survives outages:** explorer or database down for 25 s, no record lost.
- **Evidence first:** 115 automated tests (SQLite + PostgreSQL), live smoke test, every Hornet behaviour backed by a saved raw response.

TangleLens IOTA Advanced Explorer for Eclipse aeriOS Hornet **HORNET** reachable, latest milestone **15012**

Search message text: Block ID or prefix: Tag: From (UTC): To (UTC): Source: Trace:

2 open alerts: critical events or integrity failures

Status	Tag and source	Submitted (UTC)	Block
Confirmed	self.requestration	2026-10-07 12:14:46	@e441968f... b2c6e2
Content mismatch	trust.score aeriOS/trust-manager - trace domain-1-ie-2-151445	2026-10-07 12:14:46	@x26b2ef33... 7d269d
Confirmed	trust.security.alert aeriOS/self-security - trace domain-1-ie-2-151445	2026-10-07 12:14:46	@x548c1228... bb7b51
Confirmed	trust.score aeriOS/trust-manager - trace domain-1-ie-1-151445	2026-10-07 12:14:46	@xafa00f52... 7d32c3
Confirmed	trust.reliability aeriOS/self-awareness - trace domain-1-ie-1-151445	2026-10-07 12:14:46	@ccdd32b2... 8a2391

1-5 of 5

What is stored here does not match the Tangle.

decoded message in the database differs from the Tangle copy

☒ Block exists on node ☒ Block is solid ☒ Referenced by milestone

☒ Tag matches ☒ Message content matches

Stored here vs. on the Tangle

Stored in the explorer

```
{
  "domain_name": "domain-1",
  "ie": "ie-2",
  "trust_score": 0.95,
  "reliability": 0.74,
  "security": 0.52,
  "reputation": 0.7,
  "penalty": 0.05
}
```

On the Tangle (authoritative, fetched 2026-10-07 12:15:03 UTC)

```
{
  "domain_name": "domain-1",
  "ie": "ie-2",
  "trust_score": 0.61,
  "reliability": 0.74,
  "security": 0.52,
  "reputation": 0.7,
  "penalty": 0.05
}
```

Fields that differ: trust.score.

Details

Block ID: @x26b2ef338871f7e97d5f7ac43c2b20b46dc7cae0b10d713a087c4962c7